

Square-Clock Monotonicity and Finite-Clock Nonattainment for Phasewise Chowla-Free Memory

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Abstract

We close the finite-clock attainment question left open by RH-379, without enlarging its factor class. A clock q is fixed before $N \rightarrow \infty$; the admissible objects are universally distance-two-safe phasewise lag-two tables, and their interpolation coefficient $c_{11}(r)$ is zero at every phase. At the square clocks

$$q_y = 4 \prod_{i \leq y} p_i^2, \quad A_y = \prod_{i \leq y} (p_i^2 - 1), \quad D_y = \prod_{i \leq y} (p_i^2 - 2),$$

let $R_\ell^{(y)}$ count positive runs of length ℓ in the odd support word, and put

$$\mathcal{E}_y = \sum_{\ell \text{ even}} R_\ell^{(y)}, \quad L_y = \sum_{\ell \text{ even}} \ell R_\ell^{(y)}, \quad M_y = \sum_{\ell \text{ odd}} (\ell - 1) R_\ell^{(y)}.$$

Deleting the new prime-square residue from replicated old runs gives the all-order identity

$$\mathcal{E}_{y+1} = (p_{y+1}^2 - 2)\mathcal{E}_y + M_y.$$

Combining it with the locked RH-374 odd-run recurrence and the exact RH-379 square-clock formula yields an exact increment. Since $L_y - 2\mathcal{E}_y = 2R_4^{(y)} + 4R_6^{(y)} + 6R_8^{(y)} \geq 6$, the sequence $G(q_y)$ is strictly increasing.

We then prove a deliberately special saturation law: if $q_y \mid Q$ and Q has exactly the same prime support, the mod-4 and mod-9 zero-weight separators split the fine addition-by-two cycles into replicated finite paths, and $G(Q) = G(q_y)$. For arbitrary fixed q , taking $Q = \text{lcm}(q, q_y)$ therefore gives

$$G(q) \leq G(q_y) < B_\infty, \quad B_\infty - G(q) \geq \frac{12}{\pi^2 A_y (p_{y+1}^2 - 1)}.$$

Thus the RH-379 all-clock supremum is not attained by any finite clock. This is not a growing-clock result, a general cyclic-cover theorem, an adaptive-capacity limit, an operator or trace construction, a zero identification, or a statement about the Riemann hypothesis.

Keywords: Möbius function; square clocks; max-plus optimization; run deletion; finite-clock nonattainment.

1 Frozen class and predecessor inputs

The object optimized here is exactly the RH-379 phasewise Chowla-free memory class [2]. We restate the boundary because every theorem below depends on it.

Definition 1.1 (Fixed-clock phasewise class). Fix a finite integer $q \geq 1$ before taking any natural-density limit. At each $r \in \mathbb{Z}/q\mathbb{Z}$, choose a table

$$f_r : \{-1, 0, 1\}^2 \longrightarrow \{-1, +1\}, \quad \epsilon_n = f_{n \bmod q}(\mu(n-2), \mu(n)).$$

The phase family is universally safe if no ternary input can produce $\epsilon_n = \epsilon_{n+2} = +1$. It belongs to the declared class only if the coefficient $c_{11}(r)$ of $\mu(n-2)\mu(n)$ in the RH-379 interpolation is zero for every phase r . Let $G(q)$ be the largest absolute fixed-clock limiting correlation over this class.

The phasewise condition is essential. If $c_{11}(r) \neq 0$ at even one phase, the limiting expansion contains a phase-weighted shift-two Möbius correlation not controlled by the frozen sources. No cancellation between different phases is assumed here.

We use three exact predecessor results. RH-374 introduces the square-clock support word, proves its run formula and odd-run recurrence, proves the persistence of a length-eight run, and defines B_∞ as the limit of its strictly increasing square-clock sequence [3]. RH-375 identifies the same constant as the all-clock supremum in the one-site class [1]. RH-379 reduces the present memory class to the three actions $0, J, I$, proves input reflection, and establishes

$$\sup_{q < \infty} G(q) = B_\infty. \quad (1)$$

It did not decide finite-clock attainment.

For reference, the exact progression weights at phase r are

$$w_r(0) = 0, \quad w_r(J) = \delta_{q,r} - \vartheta_{q,r}, \quad w_r(I) = \delta_{q,r},$$

where $\delta_{q,r}$ is the squarefree density and $\vartheta_{q,r}$ is the joint squarefree density at displacement two. Along an addition-by-two cycle, the compatibility matrix is

$$\begin{array}{c|ccc} & 0 & J & I \\ \hline 0 & 1 & 1 & 1 \\ J & 1 & 1 & 0 \\ I & 1 & 1 & 0 \end{array}. \quad (2)$$

In particular, action 0 is compatible with everything in both directions. RH-379 proves that the positive three-state optimum equals $G(q)$: input reflection realizes the negative of every limiting score, so the absolute optimum introduces no additional case.

2 Square-clock runs and a deletion recurrence

Let $3 = p_1 < p_2 < \dots$ be the odd primes and define

$$P_y = \prod_{i \leq y} p_i^2, \quad q_y = 4P_y, \quad A_y = \prod_{i \leq y} (p_i^2 - 1), \quad D_y = \prod_{i \leq y} (p_i^2 - 2). \quad (3)$$

On $\mathbb{Z}/P_y\mathbb{Z}$, set

$$w_y(k) = \mathbf{1}_{\{p_i^2 \nmid 2k+1 \text{ for every } i \leq y\}}. \quad (4)$$

The multiples of 9 cut its cyclic positive runs, so every run has length at most eight. Write $R_\ell^{(y)}$ for the number of runs of exact length ℓ . We need four statistics:

$$O_y = R_1^{(y)} + R_3^{(y)} + R_5^{(y)} + R_7^{(y)}, \quad \mathcal{E}_y = R_2^{(y)} + R_4^{(y)} + R_6^{(y)} + R_8^{(y)}, \quad (5)$$

$$L_y = 2R_2^{(y)} + 4R_4^{(y)} + 6R_6^{(y)} + 8R_8^{(y)}, \quad M_y = 2R_3^{(y)} + 4R_5^{(y)} + 6R_7^{(y)}. \quad (6)$$

Thus L_y counts sites inside even runs, while M_y is the sum of $\ell - 1$ over odd runs. These quantities must not be conflated with one another.

RH-374 gives, with

$$e_m^{(y)} = \prod_{i \leq y} \left(1 - \frac{m}{p_i^2}\right),$$

the exact finite formula

$$R_\ell^{(y)} = \begin{cases} P_y(e_\ell^{(y)} - 2e_{\ell+1}^{(y)} + e_{\ell+2}^{(y)}), & 1 \leq \ell \leq 7, \\ P_y e_8^{(y)}, & \ell = 8. \end{cases} \quad (7)$$

It also proves

$$O_{y+1} = (s-1)O_y + L_y, \quad s = p_{y+1}^2. \quad (8)$$

Lemma 2.1 (Per-run deletion ledger). *Let an old positive run have length $1 \leq \ell \leq 8$, and let $s = p_{y+1}^2$. Across the s replicated copies used to pass from w_y to w_{y+1} , the number of descendant even positive runs is*

$$\begin{cases} s-2, & \ell \text{ even}, \\ \ell-1, & \ell \text{ odd}. \end{cases} \quad (9)$$

Proof. Represent each cyclic old run as an unwrapped interval bracketed by its two old zero sites. Those zeros remain zero in every lift, so no run crosses the chosen P_y seam and no descendant can merge across either bracket. Because $(P_y, s) = 1$, each old positive site has exactly one lift at which the new prime square divides $2k+1$. If two distinct positions d, d' in the same run were deleted in the same copy, subtraction would give $2(d-d') \equiv 0 \pmod{s}$. This is impossible because $0 < 2|d-d'| \leq 14 < s$ and every new $s = p_{y+1}^2$ is at least 25. Hence an old run has $s-\ell$ untouched copies and exactly one single-site deletion at each of its ℓ positions, all in distinct copies.

Suppose first that ℓ is even. Each untouched copy contributes one even run, giving $s-\ell$. Deleting either endpoint leaves one odd run and contributes no even run. Each of the $\ell-2$ internal deletions leaves two pieces of opposite parity, exactly one of which is a positive even run. The total is $(s-\ell) + (\ell-2) = s-2$.

Now suppose that ℓ is odd. Untouched copies contribute no even run. For $\ell=1$, deleting the sole site leaves no positive piece, agreeing with $\ell-1=0$. Let $\ell \geq 3$. For a deletion at position $d \in \{0, \dots, \ell-1\}$, the two pieces have lengths d and $\ell-1-d$, hence the same parity. Each of the two distinct endpoints contributes one positive even piece, and every internal even deletion position contributes two. The total is $2 + 2(\ell-3)/2 = \ell-1$. Odd deletion positions contribute none. This proves (9). $\square$

Proposition 2.2 (Even-run recurrence). *For every $y \geq 1$,*

$$\boxed{\mathcal{E}_{y+1} = (s-2)\mathcal{E}_y + M_y, \quad s = p_{y+1}^2.} \quad (10)$$

Proof. Sum Lemma 2.1 over all old runs. Each old even run supplies $s-2$ descendant even runs. An old odd run of length ℓ supplies $\ell-1$, and the sum of those contributions is precisely M_y . Old zero sites remain zero, while new deletions only split positive runs; there is therefore no uncounted merger across old run boundaries. $\square$

3 Exact increment and strict monotonicity

The locked RH-379 square-clock formula can be written

$$G(q_y) = \frac{4}{\pi^2} + \frac{2O_y + 4\mathcal{E}_y}{A_y\pi^2} - \frac{\mathcal{E}_y}{D_y}\kappa_2. \quad (11)$$

Put

$$H_y = \kappa_2 \frac{A_y}{D_y}. \quad (12)$$

RH-379 proves the exact Euler-product inequality

$$0 < H_y < \frac{4}{\pi^2}, \quad H_y \longrightarrow \frac{4}{\pi^2}. \quad (13)$$

No finite decimal is used in this statement.

Theorem 3.1 (Exact square-clock increment). *For $s = p_{y+1}^2$,*

$$\boxed{G(q_{y+1}) - G(q_y) = \frac{2(L_y - 2\mathcal{E}_y)}{\pi^2 A_y(s-1)} + \frac{M_y}{A_y(s-1)} \left(\frac{4}{\pi^2} - H_{y+1} \right)}. \quad (14)$$

Consequently

$$G(q_{y+1}) - G(q_y) \geq \frac{12}{\pi^2 A_y(s-1)} > 0. \quad (15)$$

Thus $G(q_y)$ is strictly increasing for every $y \geq 1$.

Proof. Besides (8) and (10), the definitions give

$$A_{y+1} = (s-1)A_y, \quad D_{y+1} = (s-2)D_y.$$

Substituting all four recurrences into (11) and subtracting the y expression gives, in the exact basis $\mathbb{Q}\pi^{-2} + \mathbb{Q}\kappa_2$,

$$\frac{2(L_y - 2\mathcal{E}_y) + 4M_y}{A_y(s-1)\pi^2} - \frac{M_y}{D_y(s-2)}\kappa_2. \quad (16)$$

Since

$$\frac{H_{y+1}}{A_y(s-1)} = \frac{\kappa_2}{D_y(s-2)},$$

equation (16) is exactly (14).

The first numerator has the combinatorial identity

$$L_y - 2\mathcal{E}_y = 2R_4^{(y)} + 4R_6^{(y)} + 6R_8^{(y)}. \quad (17)$$

RH-374 proves $R_8^{(y)} \geq 1$, so this is at least six. Also $M_y \geq 0$ by definition, and the parenthesis in the second line of (14) is positive by (13). Dropping that nonnegative term yields (15). $\square$

Remark 3.2 (What is not monotone). RH-379 writes $G(q_y) = B_y + \Delta_y$ and proves $\Delta_y \rightarrow 0$. Theorem 3.1 proves monotonicity of the sum $G(q_y)$; it neither uses nor asserts monotonicity of Δ_y .

4 A separator-specific same-support theorem

The next result is not a general statement about covers of weighted cyclic graphs. Its proof uses the precise local densities and zero-weight phases of the square clocks.

Theorem 4.1 (Same-prime-support square-clock saturation). *Fix $y \geq 1$. Suppose $q_y \mid Q$ and Q has exactly the same prime divisors as q_y . Writing $Q = Rq_y$, one has*

$$\boxed{G(Q) = G(q_y)}. \quad (18)$$

Proof. Every prime in either clock occurs to exponent at least two. The exact local squarefree formula therefore says that a phase has positive δ precisely when it is nonzero modulo every supported prime square. At such a phase,

$$\delta_{q_y, r} = \frac{2}{A_y \pi^2}, \quad \delta_{Q, t} = \frac{2}{R A_y \pi^2} \quad (t \equiv r \pmod{q_y}), \quad (19)$$

and both sides are zero simultaneously otherwise. Likewise, the exact two-squarefree formula gives

$$\vartheta_{q_y, r} = \frac{\kappa_2}{2D_y}, \quad \vartheta_{Q, t} = \frac{\kappa_2}{2RD_y} \quad (t \equiv r \pmod{q_y}), \quad (20)$$

whenever neither the current nor its predecessor is forced divisible by a supported prime square, with simultaneous zero otherwise. Thus every three-state weight on a fine phase is exactly $1/R$ times the weight of its projection modulo q_y .

Addition by two has an even and an odd cycle because both clocks are even. On the even cycle, a phase divisible by 4 has $\delta = \vartheta = 0$. On the odd cycle, a phase divisible by 9 has $\delta = \vartheta = 0$. At a zero-weight phase, replace the action by 0. This cannot lower the score and, by (2), can only relax the two adjacent compatibility conditions. These mod-4 and mod-9 phases are therefore genuine separators.

Under projection modulo q_y , each fine addition-by-two cycle traverses the corresponding base cycle exactly R times. The separators cut it into R copies of the same finite path collection. Each copied path has weights scaled by $1/R$ by (19)–(20). Max-plus path optimization is positively homogeneous, so the R copies together have exactly the base optimum. Summing the even and odd components proves (18). Finally, the RH-379 input reflection transfers the positive optimum identity to the absolute value defining G . $\square$

Remark 4.2 (Scoped negative control). The prime-support hypothesis cannot be erased from this proof. For example, $36 \mid 180$, but 180 adds the prime 5, and the exact values are

$$G(36) = \frac{9}{2\pi^2} - \frac{\kappa_2}{7}, \quad G(180) = \frac{73}{16\pi^2} - \frac{25\kappa_2}{161}.$$

The inequality is exact rather than a comparison of formal coefficient pairs: with $H = \pi^2 \kappa_2 < 4$ from the locked RH-379 enclosure,

$$G(180) - G(36) = \frac{1}{\pi^2} \left(\frac{1}{16} - \frac{2H}{161} \right) > \frac{1}{\pi^2} \left(\frac{1}{16} - \frac{8}{161} \right) > 0.$$

This is a negative control against a general-multiple or general-cover reading of Theorem 4.1; it is not a counterexample to the theorem's same-support statement.

5 Finite-clock nonattainment and an explicit gap

Lemma 5.1 (Clock divisibility). *If $q \mid Q$, then $G(q) \leq G(Q)$.*

Proof. Repeat every phase table modulo Q according to its residue modulo q . The output word is unchanged for every input, so universal safety, $c_{11} = 0$, and the fixed-clock limiting score are unchanged. The Q -class can only have a larger optimum. Input reflection again identifies the positive and absolute formulations. $\square$

Theorem 5.2 (Quantitative finite-clock nonattainment). *Let $q \geq 1$ be any fixed finite clock. Choose $y \geq 1$ so that every odd prime divisor of q belongs to $\{p_1, \dots, p_y\}$. Then*

$$G(q) < B_\infty, \quad B_\infty - G(q) \geq \frac{12}{\pi^2 A_y (p_{y+1}^2 - 1)} > 0. \quad (21)$$

Consequently no fixed finite clock attains the all-clock supremum (1).

Proof. Set

$$Q = \text{lcm}(q, q_y).$$

Then $q \mid Q$ and $q_y \mid Q$. The prime support of Q is exactly that of q_y : the choice of y contains all odd divisors of q , while the factor 4 in q_y handles the prime 2. This remains true for an arbitrary 2-adic exponent in q and arbitrary exponents of its supported odd primes; those exponents merely enlarge $R = Q/q_y$. Lemma 5.1 and Theorem 4.1 give

$$G(q) \leq G(Q) = G(q_y). \quad (22)$$

RH-379 proves $G(q_j) \rightarrow B_\infty$. Theorem 3.1 makes that sequence strictly increasing, hence $G(q_y) < B_\infty$. More quantitatively, telescope the nonnegative increments:

$$B_\infty - G(q_y) = \sum_{j \geq y} (G(q_{j+1}) - G(q_j)) \geq G(q_{y+1}) - G(q_y) \geq \frac{12}{\pi^2 A_y (p_{y+1}^2 - 1)}.$$

Combining this with (22) proves (21). $\square$

The quantifiers in Theorem 5.2 are fixed-clock quantifiers. The index y may be chosen after the finite integer q is given, but both q and the auxiliary clock Q are fixed before the $N \rightarrow \infty$ limits already used to define G . There is no diagonal choice $q = q(N)$.

6 Exact certificate and reproduction protocol

The standard-library artifact is a proof companion, not a source of the all- y theorem. It implements exact rational arithmetic in the basis $u/\pi^2 + v\kappa_2$ and fails closed if the locked enclosure $3.18 < \pi^2 \kappa_2 < 3.19$ cannot decide a max-plus comparison. The release builder verifies that the sharper RH-379 rational interval lies strictly inside this coarse interval.

The certificate performs four independent finite checks:

1. it compares the Euler-product run formula with direct cyclic words for $y = 1, 2, 3$ and samples the per-run deletion ledger at $s = 25, 49, 121$;
2. it checks the recurrence and increment algebra, including the exact anchors in Table 1;
3. for nine same-support refinements it checks every fine-residue δ and ϑ scaling law, the mod-4/mod-9 separator causes, run replication, and an independent generic three-state cyclic max-plus dynamic program;
4. it locks $Q = 180$ as a new-prime negative control and records finite least-common-multiple fixtures for the exponent scope.

Table 1: Exact finite anchors. These rows reproduce identities; the all-order proofs are Theorem 3.1 and Theorem 4.1.

y	$\mathcal{E}_y$	L_y	M_y	$L_y - 2\mathcal{E}_y$	$G(q_{y+1}) - G(q_y)$
1	1	8	0	6	$\frac{1}{16\pi^2}$
2	23	160	24	114	$\frac{9}{256\pi^2} - \frac{24}{7567}\kappa_2$
3	1105	7160	1512	4950	$\frac{443}{30720\pi^2} - \frac{216}{128639}\kappa_2$

The result ledger has a recursively closed Draft 2020-12 schema. Its 24 source locks exclude the mutable project instructions and handoff file. For each locked predecessor input, the builder checks both the live SHA-256 and byte identity with the file at its declared release commit. To replay from this directory, run

```

make result
make test
make pdf
make archive

```

The semantic PDF must be byte-identical to `main.pdf`; archive verification fails on any missing member, hash mismatch, source-lock drift, or PDF mismatch.

7 Boundaries and next theorem edge

The proved result is narrow but complete: it supplies an all-order even-run recurrence, exact strict square-clock monotonicity, a separator-specific saturation theorem, finite-clock nonattainment, and an explicit positive gap within the frozen RH-379 class.

The following statements are outside the theorem.

- No table with phasewise $c_{11}(r) \neq 0$ is covered. The first missing object is a phase-weighted shift-two Möbius correlation theorem.
- Same-support saturation is not asserted for an arbitrary weighted cyclic graph, an arbitrary cover, or a multiplier that introduces a new prime.
- No monotonicity is asserted for the correction Δ_y .
- No growing clock $q(N)$ and no convergence to adaptive distance-two capacity is asserted.
- The construction supplies no canonical intrinsic dynamical spectral determinant, time-oriented scattering completion, self-adjoint generator, von Mangoldt prime-power trace identity, or completed-zeta divisor equality. Gates A–E therefore remain false/open.
- There is no Hilbert–Pólya operator, no identification of Riemann zeros, and no implication for the Riemann hypothesis.

Immediate analytic rate trigger (not proved here). There is a within-class refinement beyond nonattainment. Put

$$T_y = \sum_{p > p_y} \frac{1}{p^2 - 1}, \quad e_m = \prod_{p \text{ odd}} \left(1 - \frac{m}{p^2}\right),$$

and

$$X_\infty = \frac{2e_4 - 4e_5 + 6e_6 - 8e_7 + 10e_8}{e_1} > 0.$$

Indeed, if $E_m^{(y)} = \prod_{i \leq y} (1 - m/p_i^2)$, the locked run formula gives

$$\frac{L_y - 2\mathcal{E}_y}{A_y} = \frac{2E_4^{(y)} - 4E_5^{(y)} + 6E_6^{(y)} - 8E_7^{(y)} + 10E_8^{(y)}}{E_1^{(y)}} \longrightarrow X_\infty.$$

The same quantity is the normalized $2R_4^{(y)} + 4R_6^{(y)} + 6R_8^{(y)}$, so $X_\infty \geq 6e_8/e_1 > 0$. This identifies the proposed first-order coefficient; it does not prove the tail asymptotic. The exact reopen target is

$$B_\infty - G(q_y) = \frac{2X_\infty}{\pi^2} T_y + O(T_y^2),$$

with the successor required to show that the M_y contribution is only second order. RH-380 does not prove this rate theorem; proving the required Euler-product expansion and $O(T_y^2)$ remainder, not computing further finite rows or fitting decay, is the trigger.

First class-enlargement blocker. Extending beyond $c_{11} = 0$ still requires a genuine theorem for phase-weighted shift-two Möbius correlations. Without such an input, the unrestricted-memory route remains STOP_SCOPED.

Declarations

Data and code availability. All theorem inputs, exact code, tests, result schema, source locks, and archive manifests are contained in the repository paper directory. No private dataset is used.

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AI assistance disclosure. AI-assisted tools supported algebra checking, artifact generation, drafting, and adversarial review. Every released claim is constrained by the locked repository sources and deterministic verification artifacts; responsibility for the scoped mathematical statements remains with the research program.

References

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